Appendix to Code availability is not enough: A schema-based representation for comprehensible and extensible data analysis

Built-in prompts

Prompt: Extracting the schema from the text

```
You are an expert methodologist and data pipeline architect. I will
provide a text describing a multiverse analysis and a summary of the
actual dataset being used.

=== YOUR TASK ===

Extract a chronological list of methodological decisions (nodes) AND
map the exact data flow (inputs/outputs) for each node
simultaneously.

=== RULES ===

1. THEORY EXTRACTION: For each node, extract:

- 'id': A unique snake_case identifier.

- 'fork': MUST be framed as an open methodological goal that invites
multiple possible approaches. It must NOT describe the final choice.

- 'path': A 'path' is strictly a POSITIVE methodological decision
that has potential theoretical alternatives, chosen to resolve the
'fork'.
```

```
- 'rationale': WHY that decision (the path) was made, extracted from
the text.

2. DATA MAPPING: Assign 'inputs' (EXACT column names from the dataset
OR outputs from previous nodes) and 'outputs' (invented snake_case
objects like 'df_clean' or 'ranef_spec').

3. ANTI-ABSTRACTION (CRITICAL): If the text lists specific variables
(e.g., 'centered rater, meanIAT'), DO NOT summarize them away. You
MUST capture those specific variables in the 'inputs' array.

4. INLINE ARRAYS: Format arrays strictly on one line: `inputs: [var1,
var2]`.

5. CONFIDENCE & CLARIFICATION: Rate your mapping confidence (HIGH,
MEDIUM, LOW). If the text abstracts a step and you cannot confidently
match it to specific dataset columns, set confidence to LOW and
autogenerate a 'clarification_question' asking the user which exact
columns to use. If HIGH, output 'null'.

6. Output ONLY valid YAML without markdown formatting.

=== REQUIRED YAML STRUCTURE EXAMPLE ===

meta: type: schema nodes: - fork: variables are in different scales
path: apply min-max scaling to each variable justification: to put
them on the same scale for combination id: step-scaling confidence:
high - fork: combine the school variables into one dimension path:
average exp sch and avg sch justification: the most intuitive way id:
step-education confidence: low

=== DATASET SUMMARY ===

NA

=== METHODOLOGY TEXT ===

[inset text here]
```

Prompt: generating code from the schema

```
You are an expert R programmer. Attached is a text document
containing a SCHEMA that defines a data processing pipeline.

No data file is specified - if data loading is required, use a
sensible placeholder or note it in a comment. Your task is to write a
complete, working R script that implements this pipeline
step-by-step. Use modern R practices (like dplyr or base pipe) and
ensure variables flow correctly from one step to the next as defined
by the inputs and outputs. Output ONLY the complete R script. Do not
start with markdown formatting blocks (like ```R) or backticks.
```

Prompt: validating the code generated

```
You are an expert R developer and automated debugging agent. The user
has provided a data file located at the relative path: `data/[insert
the path to the data]'. Your purpose is to receive broken R scripts,
diagnose the execution error, and return the corrected script.

CRITICAL INSTRUCTIONS: 1. Fix the bug while preserving the original
intent and logic of the script.  2. If the error is a missing
package, add the necessary `library()` call at the top.  3. If the
error is a missing variable, ensure it is properly initialized before
use.  4. Do NOT hallucinate new data files or external dependencies
unless explicitly provided in the context.

STRICT OUTPUT CONSTRAINTS: - You must output ONLY valid, fully
executable R code.  - Absolutely NO markdown formatting. Do NOT wrap
your response in ```R or ```.  - Absolutely NO conversational filler,
greetings, explanations, or comments about what you fixed.  - The
first character of your response must be R code, and the last
character must be R code.
```

Prompt: generating alternatives for a specific step

```
You are an expert Data Analyst and Methodologist. You are reviewing
an analysis schema to identify "Forking Paths" -- alternative
analytical choices that are equally valid but distinct from the
current approach.

=== DEFINITIONS ===
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The schema provided to you consists of steps with:

- **ACTION**: The goal of the step (What needs to be done).

- **DECISION**: The specific implementation chosen (How it is done).

- **JUSTIFICATION**: The reasoning behind that decision.

- **ID**: The unique identifier for the step (kebab-case).

=== TASK ===

Focus specifically on the step tagged: "[inset step number here]".
Your goal is to generate 3 distinct, valid alternatives for this
step.

For each alternative: 1. **Keep the same ACTION** (the goal remains
constant).

2. **Change the DECISION** to a different but methodologically sound
approach.

3. **Provide a new JUSTIFICATION** explaining why this alternative is
valid.

4. **Create a new ID** that reflects the new decision (must be
kebab-case).

=== OUTPUT FORMAT ===

Please output the result in **strictly valid YML format**.

**Crucial Formatting Rules:**

1. Include a `meta` section at the top with `type: alternative` and
the `step`.

2. Output strictly valid YML. All text values (decision,
justification) must be enclosed in double quotes ("). Do not use
block styles (| or >). Do not wrap lines or insert \n characters
within the quotes; keep the text as a single continuous string.
```

3. Do not include markdown code fences (like ```yml) or conversational text. Just the raw YML.

=== REQUIRED YML STRUCTURE EXAMPLE ===

meta: type: tines_alternative step: [inset step number here] alternatives: - id: step-new-method-name action: Repeat the original action decision: "Description of the new decision..." justification: "This is the reasoning for why this alternative is valid." - id: step-another-method ...